14 An ideal gas has thermodynamic temperature T, and its molecules has root-mean-square speed c_{rms}. If the gas is heated to temperature $2T$, what is the new root-mean-square speed of the molecules?

A $\sqrt{2}c_{\text{rms}}$ **B** $2c_{\text{rms}}$ **C** $2\sqrt{2}c_{\text{rms}}$ **D** $4c_{\text{rms}}$

Ans: A

$$\frac{1}{2}mc_{\text{rms}}^2 = \frac{3}{2}kT \Rightarrow c_{\text{rms}}^2 \propto T \Rightarrow \frac{T}{2T} = \frac{c_{\text{rms}}^2}{c_{\text{rms, new}}^2} \Rightarrow c_{\text{rms, new}} = \sqrt{2}c_{\text{rms}}$$

15 An ideal gas is contained in two spherical containers X and Y of volume $2V$ and V